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Cost reduction of high-order coupled-cluster methods via active-space and orbital transformation techniques

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We discuss several techniques which have the potential to decrease the computational expenses of high-order coupled-cluster (CC) methods with a reasonable loss in accuracy. In particular, the CC singles, doubles, and triples (CCSDT) as well as the CC singles, doubles, triples, and perturbative quadruples [CCSDT(Q)] methods are considered, which are frequently used in high-precision model chemistries for the calculation of iterative triples and quadruples corrections. First, we study the possibilities for using active spaces to decrease the computational costs. In this case, an active space is defined and some indices of cluster amplitudes are restricted to be in the space. Second, the application of transformed virtual orbitals is investigated. In this framework, to reduce the computation time the dimension of the properly transformed virtual one-particle space is truncated. We have found that the orbital transformation techniques outperform the active-space approaches. Using the transformation techniques, the computational time can be reduced in average by an order of magnitude without significant loss in accuracy. It is demonstrated that high-order CC calculations are possible for considerably larger systems than before using the implemented techniques. © *2011 American Institute of Physics*. [doi:10.1063/1.3569829]

I. INTRODUCTION

In the last decade, computational quantum chemistry has developed to a stage where it is capable of providing molecular properties with quantitative accuracy. The existence of efficient electronic-structure methods and analytic derivative techniques, convergent basis-set hierarchies and basis-set extrapolation schemes, relativistic *ab initio* approaches, and post-Born–Oppenheimer corrections inspired the development of high-accuracy model chemistries[1–9] which enabled calculations with chemical (∼1 kcal/mol) or with subchemical (∼1 kJ/mol) accuracy for small molecules. The common feature of these composite schemes is that they are based on the coupled-cluster (CC) hierarchy of methods. In particular, the low-level correlation effects are taken into account via the CC singles, doubles, and perturbative triples [CCSD(T)] approach,[10] while the higher-order correlation contributions are evaluated using the CC singles, doubles, and triples (CCSDT) (Ref. 11) as well as the CC singles, doubles, triples, and perturbative quadruples [CCSDT(Q)][12–14] or the CC singles, doubles, triples, and quadruples (CCSDTQ) (Refs.15–17) methods. Unfortunately, the scaling of the latter approaches with the size of the systems is steep, which seriously hinders the extension of high-precision model chemistries toward larger molecules. For this reason, it is tempting to pursue strategies to decrease the computational expenses of high-order CC methods.

There have been several attempts to develop cost-efficient approximations to CC methods. The efforts along this line can be classified in three categories: the active-space (AS) CC approaches, the methods based on the transformation and subsequent truncation of the one-electron basis set, and the approximations using some kind of decomposition of the wave function parameters or integral lists.

The AS approach was originally published by Oliphant and Adamowicz[15] and was intended to include the static electron correlation into the CC theory using the single-reference (SR) formalism. These authors supposed that there is another important determinant in addition to the Fermi vacuum, which can be generated by double substitution from the latter. The two determinants together constitute the AS. Single and double excitations out of the secondary determinant can be simulated by higher excitations with respect to the Fermi vacuum with partially fixed indices. The fixed indices correspond to those two orbitals in which the two determinants differ. This formally multireference CC method was later generalized by Piecuch and co-workers[18,19] to more than two orbitals, and implemented for arbitrary ASs by Kállay *et al.*[20] Later Köhn and Olsen[21] developed a generalization of the AS CC approach using a more sophisticated classification of the virtual orbitals and presented promising CCSD, CCSDT, and CCSDTQ results for SR systems. The AS approach have also been implemented by Fan and Hirata[22] up to the quadruples level of theory and was applied to describe the singlet-triplet separation for triatomic hydrides. The philosophy of the AS approach has been utilized in the framework of the equation of motion CC theory by Kowalski and Piecuch,[23,24] and Kowalski *et al.*[25] These methods were applied to describe the electronic state of the Be_3 cluster[25] and excited, ionized, and electron-attached states of small molecules.[26]

Though the AS approach was originally designed for systems with pronounced multireference character, it can also be

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regarded as a selection scheme for higher-than-double excitations, and thus as a cost-reduction technique for high-order SR CC methods. This aspect was already pointed out in the literature;[19,21] however, the performance of the AS approach has not been systematically tested in the context of high-accuracy calculations.

In the second category of cost-reduction techniques the set of virtual orbitals is divided into active and inactive subspaces, and the inactive orbitals are dropped. Prior to that, utilizing the rotational invariance of SR CC methods, a transformation of the virtual orbitals is performed to minimize the error coming from the neglect of inactive virtual orbitals in the subsequent correlated calculations.

The separation of the important part of the virtual space is frequently performed by the application of natural orbitals (NOs).[27–29] The eigenvalues of the one-particle density matrix obtained from a wave function ansatz can be considered as the population of the corresponding orbitals, i.e., the NOs. Using this simple physical interpretation the removal of the weakly populated NOs leads to a simple method to reduce the one-particle basis set. A plausible choice for the construction of the truncated virtual space for the purpose of CC calculations is to use the NOs obtained with the first-order Møller–Plesset (MP) wave function.[30–33]

In several approaches, a functional of the orbital rotation parameters in the virtual space is defined to maximize (or minimize) the value of some physical quantity. The corresponding modified orbitals are often referred to as optimized virtual orbitals (OVOs). Adamowicz *et al.*[34,35] minimized a modified second-order Hylleraas-functional with respect to the parameters of the rotation among the active and inactive orbitals. Later Sosa *et al.*[32] showed that the Hylleraas-functional-based approach gives more accurate results for the CCSD[T] energy[36] than the simple truncation based on MP2 NOs. For the applications where the OVOs serve the cost-reduction of high-order CC calculations, the approach of Neogrády, Pitoňák, and Urban[37] is more attractive. In their study two functionals were proposed, one of them is based on the CC energy expression, while the other makes use of the overlap of the CC wave functions defined over the full and the active virtual spaces, and it was demonstrated in several test calculations that the overlap functional gives more accurate results for the CCSD and CCSD(T) energies than the energy functional.

We note that the error introduced by the above approximations can be significantly reduced via a simple correction originally suggested by Adamowicz and Bartlett,[34] who evaluated the CCSD(T) energy as a sum of the MP2 energy calculated with the full basis set and the difference of the CCSD(T) and MP2 energies obtained with the truncated virtual space. It was demonstrated in their study and later by Taube and Bartlett[38] and Pitoňak *et al.*[39] that the correction significantly improves the performance of the methods due to error compensation.

We should also mention the so-called multiple basis set approach of Klopper and co-workers,[40] which can be applied with any of the virtual orbital optimization methods discussed above to accelerate the calculations of the perturbative triples correction in the CCSD(T) method. In this approach, the CCSD equations are solved with the original basis set, then the singles and doubles amplitudes are projected into the reduced space, where the triples correction is calculated.

The methods using different decomposition techniques are out of the scope of the present study; thus, the corresponding approaches are only briefly mentioned here. Reviving an old idea of Beebe and Linderberg,[41] Koch *et al.* used the Cholesky decomposition to factorize the two-electron integral list and significantly decreased the computational expenses of *ab initio* methods including CC approaches.[42,43] Koch and Sánchez de Merás[44] also investigated the Cholesky-decomposition of the CCSD(T) energy denominators which allows the factorization of CCSD(T) energy diagrams to obtain a reduced-scaling algorithm. Finally, Kinoshita, Hino, and Bartlett[45,46] applied the singular value decomposition (SVD) to the matrix of cluster amplitudes to develop cost-effective iterative CC methods.

In this account, we examine if the above-mentioned scaling-reduction techniques based on the AS approximations or the truncation of the one-electron basis can be applied to high-order CC methods. Several numerical studies have been already published about the efficiency of these approaches for the calculation of CCSD and CCSD(T) energies;[21,32,37,39,47,48] thus, the latter are not considered here. The structure of the paper is as follows. In Sec. II, we discuss the acceptable extent of the error in the approximations and summarize the theory of the considered techniques. In Sec. III, the results of our benchmark calculations are discussed to evaluate the cost-effectiveness and accuracy of the various approaches. To confirm the *raison d'etre* of the computation time reduction techniques large-scale CCSDT and CCSDT(Q) calculations are also presented for bigger molecules.

II. THEORY

A. Error of the approximations

The principal aim of this paper is to develop reduced-cost CCSDT and CCSDT(Q) methods using various computation time reduction techniques, which enable the cheaper evaluation of the iterative triples and perturbative quadruples contributions required in high-accuracy model chemistries. That is, we intend to compute

$$\Delta E_{\mathrm{CCSDT}} = E_{\mathrm{CCSDT}} - E_{\mathrm{CCSD(T)}}, \tag{1}$$

$$\Delta E_{\mathrm{CCSDT(Q)}} = E_{\mathrm{CCSDT(Q)}} - E_{\mathrm{CCSDT}}, \tag{2}$$

quantities at a lower cost with an acceptable loss of accuracy, where $E_{\mathrm{CCSD(T)}}$, E_{CCSDT}, and $E_{\mathrm{CCSDT(Q)}}$ are the total energies computed with the corresponding methods.

The error should be kept below a certain threshold to be able to calculate the energy with subchemical accuracy (1 kJ/mol $\approx$ 0.4 mE$_h$), which is desirable in high-accuracy model chemistries. The intrinsic error of any quantum chemical method obviously increases with growing system size, and it is hardly possible to keep this error constant. However, in most of the chemical processes only a small part of a large molecule, just a few chemical bonds are affected. For this

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reason, we aim at an accuracy of 1 kJ/mol per a chemical bond, and not for the entire molecule. For a diatomic molecule, the quadruples contribution, approximated here by $\Delta E_{\mathrm{CCSDT(Q)}}$, is usually in the 1–2 mE_h ($\approx$ 2–5 kJ/mol) range, while the iterative triples contribution ($\Delta E_{\mathrm{CCSDT}}$) is usually much smaller, typically 0.1–0.7 mE_h ($\approx$ 0.3–1.8 kJ/mol).[4,6,50] Thus, in general, we consider the results acceptable if the error of the $\Delta E_{\mathrm{CCSDT}}$ and $\Delta E_{\mathrm{CCSDT(Q)}}$ increments is less than 50 and 15%, respectively. Nevertheless, it is important to note that the absolute errors of the contributions should also be taken into account, especially for the T–(T) increment, when judging the performance of the cost-reduction techniques. The correlation contributions are sometimes very small, and we can tolerate larger relative errors, which still keep the absolute error per chemical bond far below 1 kJ/mol.

B. Active space approach

The AS CC approach introduced by Adamowicz and Piecuch[15,18] was originally developed to include static and dynamic correlation into the SR CC theory by the application of certain higher-order excitations. The molecular orbital (MO) basis is divided into active and inactive subspaces, and a complete active space (CAS) is defined by distributing the active electrons among the active orbitals.[20] One determinant of the CAS space is chosen as a Fermi-vacuum. To describe the dynamic correlation of the active space, higher excitations are included in the cluster operator but some of the indices of the higher-order cluster amplitudes are restricted to be active. For instance, at the CCSD level T_3 amplitudes can be added with the form t_{Ijk}^{Abc}, t_{IJk}^{ABc}, and t_{IJK}^{ABC}, where b, c and i, j are inactive occupied and virtual orbitals, and A,B,C, and I,J,K denote occupied and virtual active orbitals, respectively. This formally SR approach can be successfully applied to describe particular multireference problems. In a truly multireference case, however, this framework has a drawback that more than one determinant can be chosen as the principal determinant, i.e., the formal Fermi-vacuum, and the result depends on this choice.

In the single reference case, the selection of the Fermi-vacuum is unambiguous, and the AS CC approach can be used as a truncation scheme for higher excitations to obtain the major part of the dynamical correlation at a lower computational price. In this study, we used the AS CC method to truncate only the highest-order amplitude, that is, the T_3 amplitudes for the calculation of the T contribution, and the T_4 amplitudes in the case of $\Delta E_{\mathrm{CCSDT(Q)}}$. Further, only the virtual space was restricted, i.e., all the occupied orbitals were considered active. The virtual orbitals were included in the active space on the basis of increasing orbital energies. The speedup which can be gained in this way is simply $(n_v/n_{av})^m$, where m is the number of active virtual indices of the highest order cluster amplitude, n_v and n_{av} are the number of virtual and active virtual orbitals, respectively.

C. MP2 frozen natural orbitals

Diagonalizing the virtual block of a one-particle density matrix and retaining only the virtual orbitals with high population result in a reduced-size basis set called the frozen natural orbital (FNO) basis set, which hopefully contains the essential part of the virtual space. Since the computational cost of the CCSDT and CCSDT(Q) methods scales as the fifth power of n_v, the reduction of the virtual space implies a speedup which is proportional to $(n_v/n_{av})^5$.

In the CC case, the simplest approximation of the wave function is the first-order MP wave function. In our study, the virtual part of the density matrix constructed from the first-order MP wave function was diagonalized to obtain the virtual NOs, and the NOs with low population were neglected. Since the working equations for perturbative CC approaches such as CCSD(T) and CCSDT(Q) were derived for a one-electron basis where the Fock-matrix is diagonal, we constructed the canonical orbitals in the reduced one-particle space. This approach will be referred to as MP2 FNO.[32]

To ensure the error cancellation, we used the same reduced MP2 FNO basis set for the calculation of both terms at the right hand side of Eqs. (1) and (2); thus, no additional corrections are needed as in the case of the CCSD and CCSD(T) methods.[34,39]

For the truncation of the virtual NO space there exist several schemes. In the simplest case, one defines an appropriate threshold for the NO occupation numbers and drops the NOs below this threshold. Another criterion has recently been introduced by Landau *et al.*,[33] which was also adopted in our work. In this scheme, for each NO the occupation number of that particular NO and all the NOs with larger occupation numbers are summed up. This cumulative occupation number is divided by the number of electrons; thus, the resulting number—hereafter referred to as ε—lies between 0 and 1. Only those virtual orbitals are retained for which ε is smaller then a threshold, e.g., 0.9. This algorithm results in relative errors for various properties which are less sensitive to the properties of the studied systems and the employed basis sets.

D. Optimized virtual orbitals

Beyond the MP2 FNO method, one of the most promising ways for the optimization of the virtual orbitals was introduced by Neogrády *et al.*,[37] which is also followed in this study. An overlap functional for the CCSD wave functions of the full and the reduced virtual spaces is defined. If the CCSD wave function is truncated at the linear term, the functional can be written as

$$L_{\mathrm{overlap}} = \langle \Psi | \hat{T}_{\mathrm{CCSD}} | \hat{T}^*_{\mathrm{CCSD}}(\mathbf{U}) | \Psi \rangle, \tag{3}$$

where Ψ is the Hartree-Fock (HF) wave function and

$$\hat{T}_{\mathrm{CCSD}} = \sum_{ia} t_i^a a^+ i + \frac{1}{4} \sum_{ijab} t_{ij}^{ab} a^+ b^+ ji, \tag{4}$$

$$\hat{T}^*_{\mathrm{CCSD}}(\mathbf{U}) = \sum_{ia^*} t_i^{a^*} a^{*+} i + \frac{1}{4} \sum_{ija^*b^*} t_{ij}^{a^*b^*} a^{*+} b^{*+} ji,$$

with t_i^a and t_{ij}^{ab} as the single and double excitation amplitudes, i and j as occupied orbitals, a^* and b^* as the active virtual orbitals, and a and b as general virtual orbitals. In operator $\hat{T}^*_{\mathrm{CCSD}}$, the orbitals depend on the rotational matrix $\mathbf{U}$, where $\mathbf{U}$ is a unitary transformation of the initial HF virtual orbitals.

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Operator $\hat{T}_{\mathrm{CCSD}}$ is considered invariant to the rotation of the virtual orbitals, i.e., for a

$$|a'\rangle = \sum_a U_{aa'}|a\rangle, \tag{5}$$

unitary transformation of the virtual orbitals the amplitudes are supposed to change as

$$t'^{a}_{i} = \sum_b U_{ab} t^{b}_{i}, \quad t'^{ab}_{ij} = \sum_{cd} U_{ac} U_{bd} t^{cd}_{ij}. \tag{6}$$

Operator $\hat{T}^*$ is still invariant to the rotation of the active virtual orbitals and is not affected by the rotation of inactive virtuals; thus, to maximize the overlap functional, Eq. (3), those elements of **U** have to be optimized which can mix the active and inactive virtuals.

Since the working equations derived from the overlap functional have not been published explicitly, in the Appendix we give a short review of the derivation and present the formulas we have implemented.

In Ref. 37, a further approximation was also introduced. In the overlap functional, the first-order approximation,

$$^{[1]}t^{a^*b^*}_{ij} = \frac{\langle ij||a^*b^*\rangle}{\epsilon_i + \epsilon_j - \epsilon_{a^*} - \epsilon_{b^*}}, \tag{7}$$

for the doubles amplitudes t^{ab}_{ij} was used instead of the converged ones. It was demonstrated that this approximation does not lead to notable loss of accuracy in the energy. In our study, we also introduced this approximation.

We note again that additional corrections like those introduced for the CCSD and CCSD(T) methods[34,39] are not needed because we calculate energy differences [cf. Eqs. (1) and (2)], and the error compensation is automatic.

We also note that the AS CC approach could, in principle, be combined with the orbital transformation techniques, i.e., with the MP2 FNO or OVOs approaches. As Köhn and Olsen[21] showed improving the active orbitals leads to better results than those obtained applying the HF canonical orbitals in the AS CC method. In this paper, however, we also consider perturbative CC approaches such as CCSD(T) (it is needed for the triples contribution) and CCSDT(Q), which presuppose the application of canonical orbitals instead of an optimized basis; thus, we did not explore this way. We hope that the larger error of the full AS CC energies coming from the less appropriate choice of the basis set does not strongly affect the energy differences in Eqs. (1) and (2) since both terms on the right hand side of the expressions are computed at the same level of approximation.

III. RESULTS AND DISCUSSION

A. Computational details

To assess the performance of the AS CC, MP2 FNO, and OVOs techniques, numerous test calculations were carried out for small and medium-sized molecules. The results are also compared to the efficiency of the simple elimination of canonical HF MOs on the basis of orbital energies. All calculations were performed using the MRCC suite of quantum chemistry programs.[51] In the AS CC calculations, we used the implementation of the AS CC approach as described in Ref. 20, while for the MP2 FNO and OVOs calculations additional modules have been developed. In the case of MP2 FNO, the approximate MP2 density matrix has been coded, and a four-index integral transformation routine has been written. Connected to the OVOs method, we have also implemented the self-consistent iteration process discussed in the Appendix. Canonical HF MO integrals were taken from the CFOUR program package.[52]

As we pointed out in Sec. II, our aim was to monitor the performance of the cost-reduction techniques for the iterative triples and quadruples CC contributions for the purpose of high-accuracy calculations. Since these contributions are usually evaluated with Dunning's correlation consistent polarized triple- (cc-pVTZ) and double-ζ (cc-pVDZ) basis sets,[53] respectively, in high-precision model chemistries,[1–9] we also chose these basis sets. In our calculations, the core electrons were kept frozen, and the geometries optimized at the CCSD(T) level of theory with the cc-pVQZ basis set were used. As we discussed in Sec. II, to ensure the required accuracy the results with relative errors of less than 50 and 15% are considered acceptable for the iterative triples and quadruples corrections, respectively.

B. Comparison of the cost reduction techniques

First, we carried out a systematic study for the 30 atoms and molecules of the HEAT test set.[4] We have found that our cost-reduction approaches are efficient for molecules containing at least two heavy atoms and one hydrogen atom, but for smaller molecules, due to the small number of basis functions and electrons, they only give reliable results if the truncation of the virtual space is not significant. Here, only two representative examples are selected, the carbon-dioxide and hydrogen-peroxide molecules, to present the behavior of the approaches. The selection of these molecules is justified because the contribution of dynamic correlation, and consequently that of triple and quadruple excitations to the total energies, is significant. Furthermore, CO_2 and H_2O_2 represent two prototypical classes of molecules, those where the iterative triples contribution is considerable and very small, respectively. In addition to the small systems, we also carried out calculations for two larger molecules, butadiene and benzene, which were previously considered in high-precision calculations[13,54] and were considered as the upper limit for such studies.

The results for the $\Delta E_{\mathrm{CCSDT}}$ contribution in the case of the CO_2 molecule with the cc-pVTZ basis set are shown in Fig. 1, where the relative deviations from the original value, 0.209 mE_h, and the speedup of the approximate CCSDT calculations with respect to the full CCSDT calculation are displayed. In Fig. 1 and in the following the abbreviations "no inactive labels," "one inactive label," "two inactive labels," and "three inactive labels" will refer to the AS CC method where the amplitudes of the highest excitations, that is, the triple excitations in the case of the triples contribution and quadruple excitations in the case of the quadruples contributions are restricted to contain no, at most one, at most two, and at most three inactive virtual labels, respectively. The full

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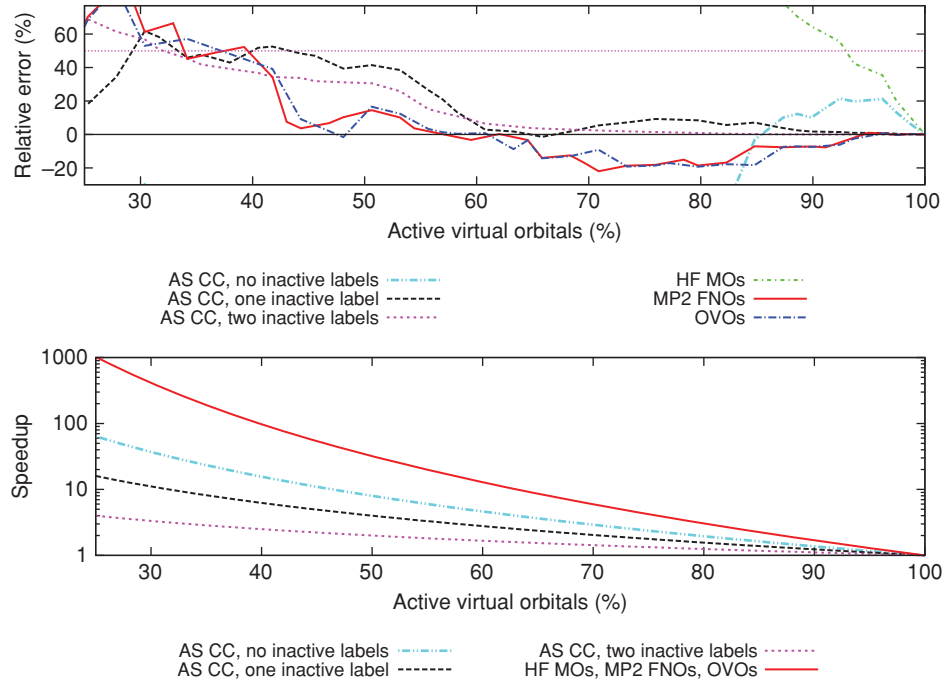

FIG. 1. Convergence of various cost-reduction techniques for the iterative triples contribution of the carbon-dioxide molecule with the cc-pVTZ basis set.

virtual space contains 79 MOs. The elimination of the high-lying HF MOs or the application of the AS CC approach without inactive labels on the T_3 amplitudes give poor convergence. The restriction of one or two labels in the AS CC calculations to be active leads to significantly better convergence giving acceptable results if more than 40% of the virtual orbitals are active. Surprisingly, the convergence is only slightly faster if two labels are allowed to be inactive. Considering the speedup, when 40% of the orbitals are active, the AS CCSDT method with one or two active labels is about 3 or 6 times faster than the parent CCSDT calculation, respectively. The OVOs and MP2 FNO methods have similar convergence behavior, showing poor convergence. Again, about 40% of the virtual space is necessary for a relative error of 50%, while the results oscillate in the 20% error interval if more than 50% of the optimized virtual orbitals are retained. The speedup at the 40% truncation is roughly a factor of 100. In absolute terms the iterative triples correction itself is relatively small, ≈0.5 kJ/mol. Consequently, a relative error of 50% means a tolerable absolute error of about 0.3 kJ/mol.

Figure 2 shows the results of the calculations for the quadruples contribution of the CO_2 molecule in the cc-pVDZ basis. The (Q) contribution in the full basis is −2.072 mE_h, which is an order of magnitude larger than the triples contribution, but it is interesting to note that the convergence is faster for CCSDT(Q) than for CCSDT for all the investigated methods. Using HF MOs or the AS CC method without inactive orbitals, one can only get satisfactory results if more than 80% of the virtual orbitals are retained, while the AS CC method with one inactive orbital on quadruple excitation amplitudes requires about 60% of the virtual orbitals for an acceptable energy. The performance of the AS CC method with two inactive orbitals and the MP2 FNO or OVOs approaches is rather similar. For these methods 40% of the virtuals are enough to keep the error in the acceptable range. The theoretical speedup at this point is close to a factor of 100 for the MP2 FNO and OVOs methods but it is only about 7 for the AS CC scheme. For this example, the application of the AS CC method with three inactive orbitals is also efficient giving good results even if few active orbitals are used, but the speedup is negligible.

The second example is the hydrogen-peroxide molecule (see Figs. 3 and 4). The T–(T) contribution for this system is very small, 0.033 mE_h. The only method which shows satisfactory convergence is the AS CC approach with two inactive labels but the speedup is rather small, less than a factor of 2. The orbital transformation methods are only able to estimate the CCSDT energy within the 50% error bar if more then 75% of the virtual orbitals is retained, but at least they can give the correct magnitude of the contribution if more than 50% of the virtuals are considered. Nevertheless, the iterative triples contribution is so small that a larger error bar or the complete elimination of this contribution would also be acceptable.

The (Q) contribution for hydrogen-peroxide in the cc-pVDZ basis is −1.414 mE_h. The performance of the various methods are similar to that for CO_2. The results using the HF MOs or the AS CC approximation with four active (zero inactive) labels are only acceptable if 75% of the virtual space is active. In the AS CC framework, increasing the number of the inactive labels results in faster convergence, i.e., using one, two, and three active labels reduces the required number of active orbitals to 45, 25, and 15%, respectively. For these truncations, the measured speedups are similar, less than a factor of 5. The MP2 FNO and OVOs methods exhibit similar convergence behavior. With these approaches one needs 25% of the virtuals for the desired accuracy. At this point, the

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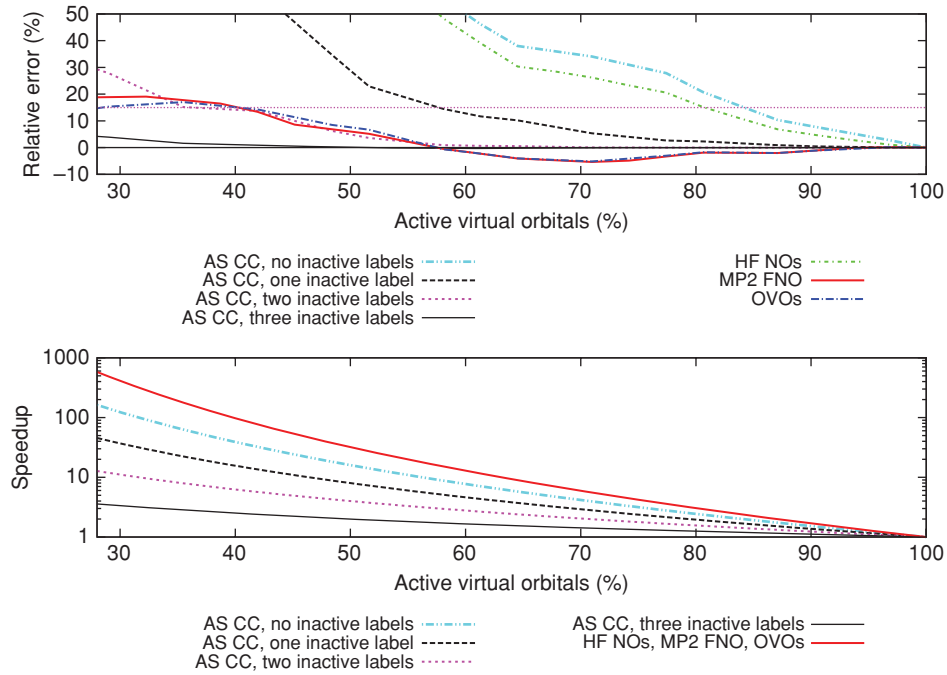

FIG. 2. Convergence of various cost-reduction techniques for the quadruples contribution of the carbon-dioxide molecule with the cc-pVDZ basis set.

measured speedup is about 500, which is significantly larger than the speedups obtained with the AS CC methods.

Since our test calculations for the smaller molecules revealed that the performance of the MP2 FNO and OVOs methods are superior to that for the AS CC approaches, AS CC calculations were not performed for the butadiene and benzene molecules, and in the following only the former methods will be considered.

The MP2 FNO and OVOs CCSDT results for the butadiene molecule (see Fig. 5) show that 35% of the 189 virtual orbitals is required for a good approximation of the -0.471 mE$_h$ triples contribution, and the speedup gained in this way is more than a factor of 100. Again, the iterative T increment is rather small, and its absolute error is significantly lower than 1 kJ/mol. The (Q) contribution with the full cc-pVDZ basis set containing 71 virtual orbitals is -1.545 mE$_h$. Figure 6 shows the MP2 FNO and OVOs results, where one can see that the (Q) contribution can be adequately estimated retaining only 30% of the virtual space. At this point, the gained speedup is in the order of several hundred. It is

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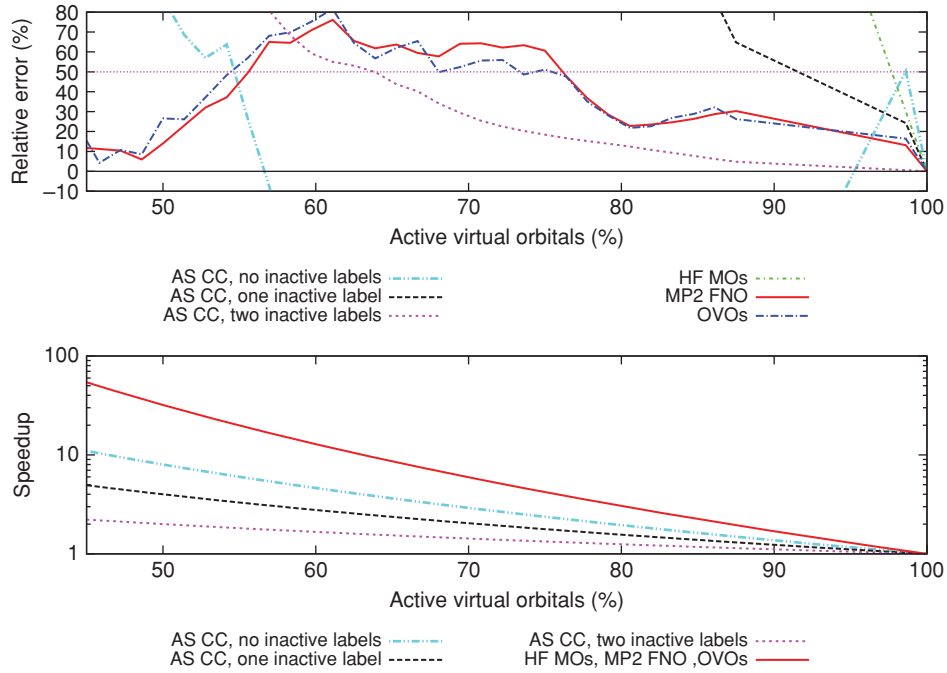

FIG. 3. Convergence of various cost-reduction techniques for the iterative triples contribution of the hydrogen-peroxide molecule with the cc-pVTZ basis set.

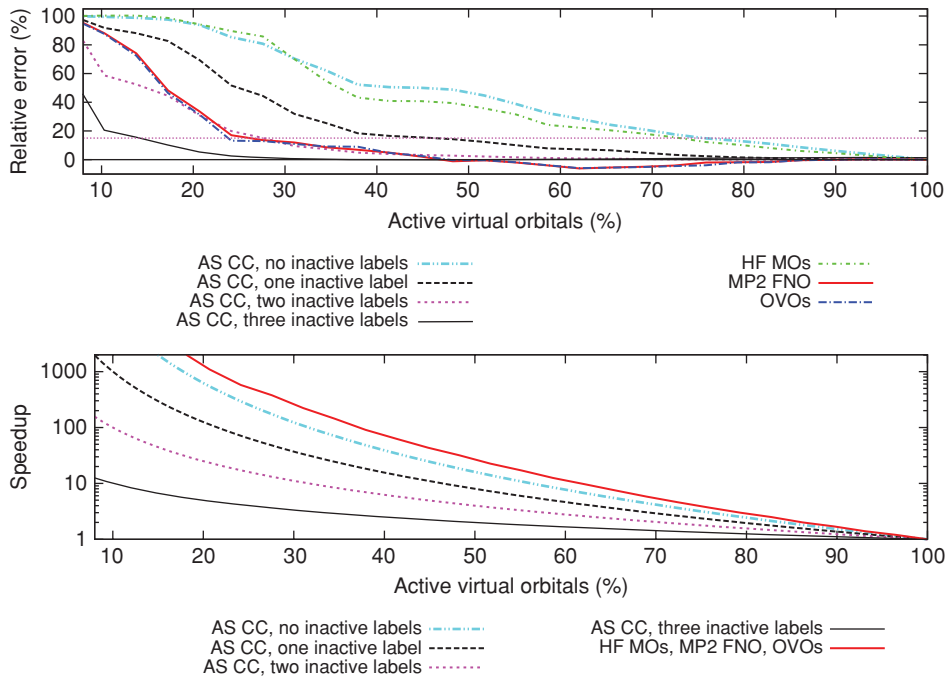

FIG. 4. Convergence of various cost-reduction techniques for the quadruples contribution of the hydrogen-peroxide molecule with the cc-pVDZ basis set.

important to note here again that there is no significant difference between the MP2 FNO and OVOs results.

For the benzene molecule, the cc-pVTZ basis set contains 258 basis functions and the T–(T) contribution is 0.598 mE$_h$, while the cc-pVDZ basis includes 108 basis functions and the (Q) contribution amounts to 2.703 mE$_h$. The results can be found in Figs. 7 and 8. For this system, similar conclusions can be drawn as for butadiene, that is, about 50 and 35% of the virtual space is necessary for satisfactory results, and thereby the calculation time is reduced roughly by two orders of magnitude for both the triples and quadruples contributions.

As we emphasized in Sec. II, when we apply the AS CC methods to the (Q) contribution, only some of the labels of the T_4 amplitudes are restricted to be active, but no restrictions are imposed on the triple excitations. We note, however, that it would also be reasonable to solve the CCSDT equations with the same number of active labels on the T_3 amplitudes as it is done for the T_4 amplitudes at the evaluation of the (Q) correction. One would hope that the error introduced in this way

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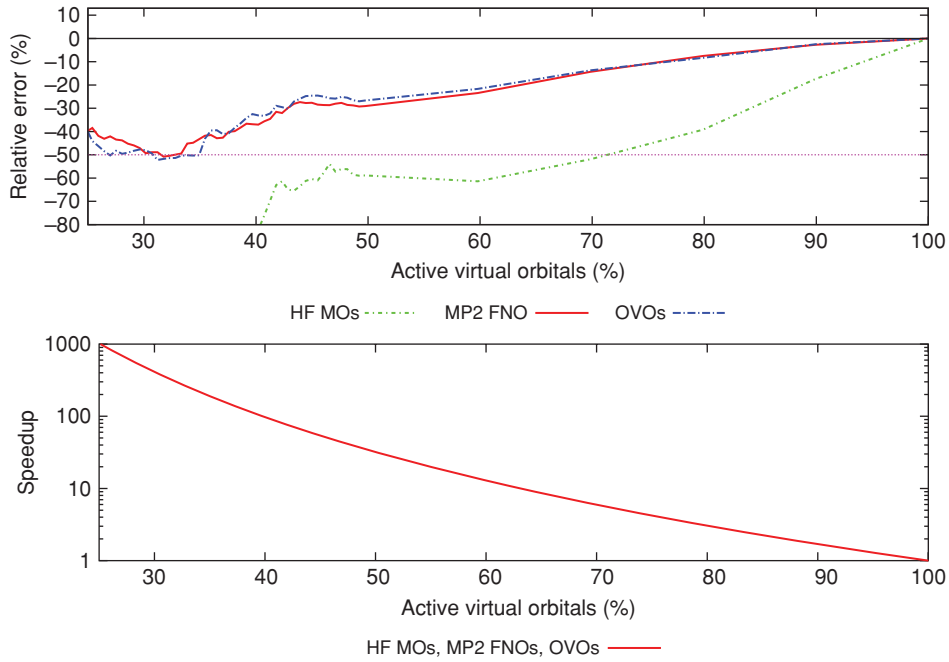

FIG. 5. Convergence of various cost-reduction techniques for the iterative triples contribution of the butadiene molecule with the cc-pVTZ basis set.

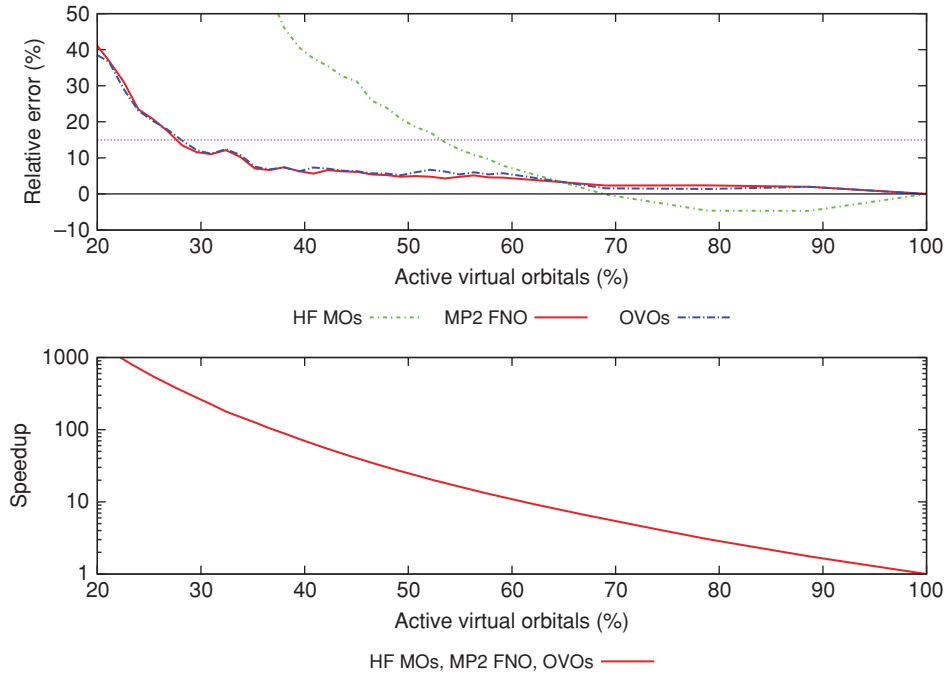

FIG. 6. Convergence of various cost-reduction techniques for the quadruples contribution of the butadiene molecule with the cc-pVDZ basis set.

to the CCSDT energy may approximately cancel the error of the CCSDT(Q) energy; thus, the error of the (Q) contribution may be smaller. We have also tested this approach and have found only a negligible improvement with respect to our original scheme, where the CCSDT energy is determined in the full virtual space.

While we are mainly interested in the approximate evaluation of the $\Delta E_{\mathrm{CCSDT}}$ and $\Delta E_{\mathrm{CCSDT(Q)}}$ increments, the former is obviously rather small, and one may ask why the two contributions are not computed together, i.e., why the post-(T) corrections are not evaluated as the difference of CCSDT(Q) and CCSD(T) energies. The experience shows that the $\Delta E_{\mathrm{CCSDT}}$ contribution converges slowly with the basis set size,[49] and at least triple-ζ quality basis sets are required for reliable results. Since CCSDT(Q) calculations are not feasible for bigger molecules with triple-ζ basis sets, the joint computation of the $\Delta E_{\mathrm{CCSDT(Q)}}$ and $\Delta E_{\mathrm{CCSDT}}$ corrections would involve that the latter is also evaluated with double-ζ basis sets.

As a conclusion of the above examples, we can say that the AS CC, MP2 FNO, and OVOs methods show significant

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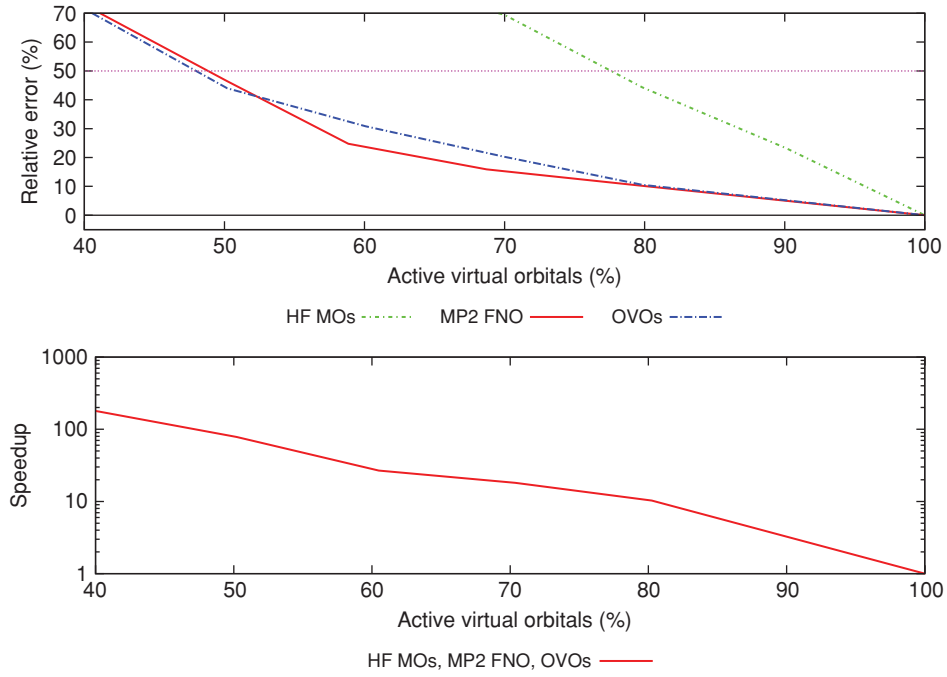

FIG. 7. Convergence of various cost-reduction techniques for the iterative triples contribution of the benzene molecule with the cc-pVTZ basis set.

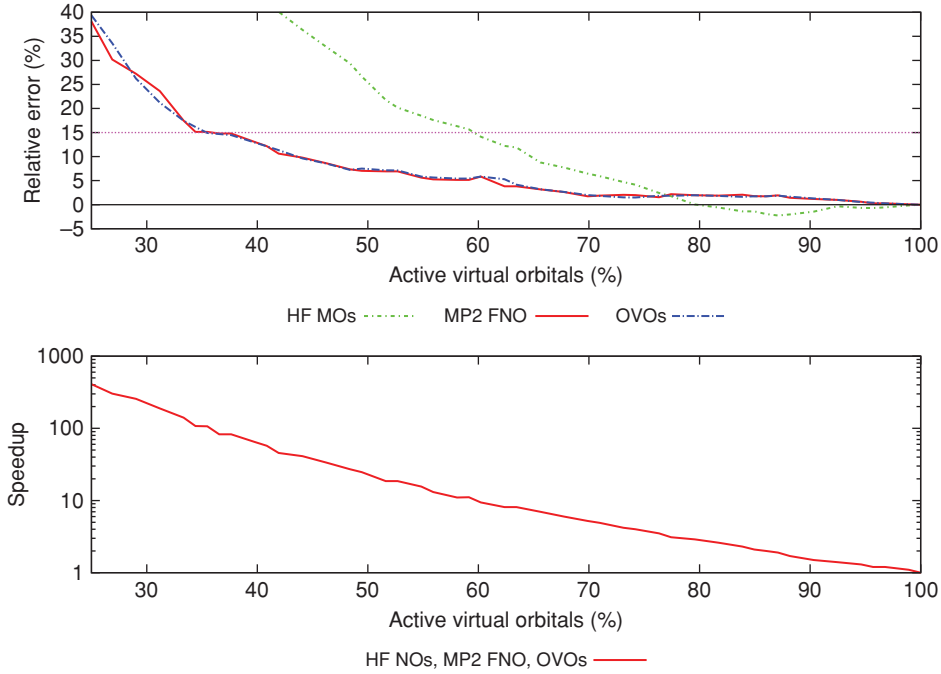

FIG. 8. Convergence of various cost-reduction techniques for the quadruples contribution of the benzene molecule with the cc-pVDZ basis set.

improvement with respect to the approach based on the simple elimination of HF canonical orbitals. The AS CC method with only one label restricted to be active converges rapidly with the size of virtual space but the gain in the speed is generally small. Application of more than one active orbitals does not result in a convergence which is faster than that for the MP2 FNO and OVOs methods, and the speedup for a given precision is also smaller. On the other hand, the MP2 FNO and OVOs methods ensure fast convergence for the (Q) corrections, and the speedup can be in the order of several hundred at the expense of a 15% loss in the accuracy. It is important to note that the approximate (Q) corrections approach the exact value from below for all the test systems, i.e., the absolute value of the approximate (Q) corrections is smaller. Since the CCSDT(Q) method almost exclusively overestimates the contribution of quadruple excitations by 10 to 20%,[12,13] the approximate (Q) corrections are probably closer to the quadruples contributions obtained with the full CCSDTQ method.

The conclusions regarding the behavior of the T–(T) increments are less obvious. The above results reveal that the convergence of the iterative triples contribution with respect to the active space size is relatively slow. Provided that the T contribution is not negligible the OVOs or MP2 FNO methods require up to 50% of the virtual space to obtain results with less than 50% relative error, which implies at least a 30-times speedup in computational time. The AS CC method with one active label requires somewhat smaller active space than the OVOs or MP2 FNO approaches for satisfactory results, but the improvement achieved in the computational expenses of the CCSDT calculations is not significant. In the case if the contribution is very small, only 10 to 30% of the virtuals should be frozen to keep the relative error below the limit; however, the absolute error remains below 1 kJ/mol even if the relative error is larger than 50%.

It is interesting to observe that the MP2 FNO and the more sophisticated OVOs methods give rather similar results for both the iterative triples and quadruples contributions. Since the MP2 FNO approach is somewhat more appealing because of its simplicity, in the following we will focus on this approach.

C. Performance of the MP2 FNO approach for correlation contributions

The above test calculations show that usually a considerable part of the virtual space can be ignored at the calculation of the triples and quadruples contributions; however, the actual fraction of virtual orbitals that are required for a given accuracy may somewhat vary. Thus, it would be helpful to define some criterion for the truncation which ensures that the relative errors of the approximate correlation contributions lie below the desired limits. Of course, such a criterion should be conservative enough to guarantee the uniform accuracy of the methods for a wider range of molecules. This means, however, that the criterion may be too conservative for particular systems and result in higher accuracy than needed but, at the same time, in lower speedups. To find the optimal truncation and to gain larger speedup, one can try to utilize *a priori* information about the investigated molecule like previous calculations on similar systems or using several cutoff parameters with a proper extrapolation technique.[33,55]

To find a general criterion, here we invoke the approach of Landau *et al.*[33] outlined in Sec. II C and seek a sufficient threshold for the sum of the NO occupations (ε) of MP2 NOs. To that end, we carried out benchmark calculations for a representative test set of molecules. Recently, we computed thermodynamic functions for numerous molecules including chloro-fluoro-carbons,[56] formaldehyde derivatives,[57]

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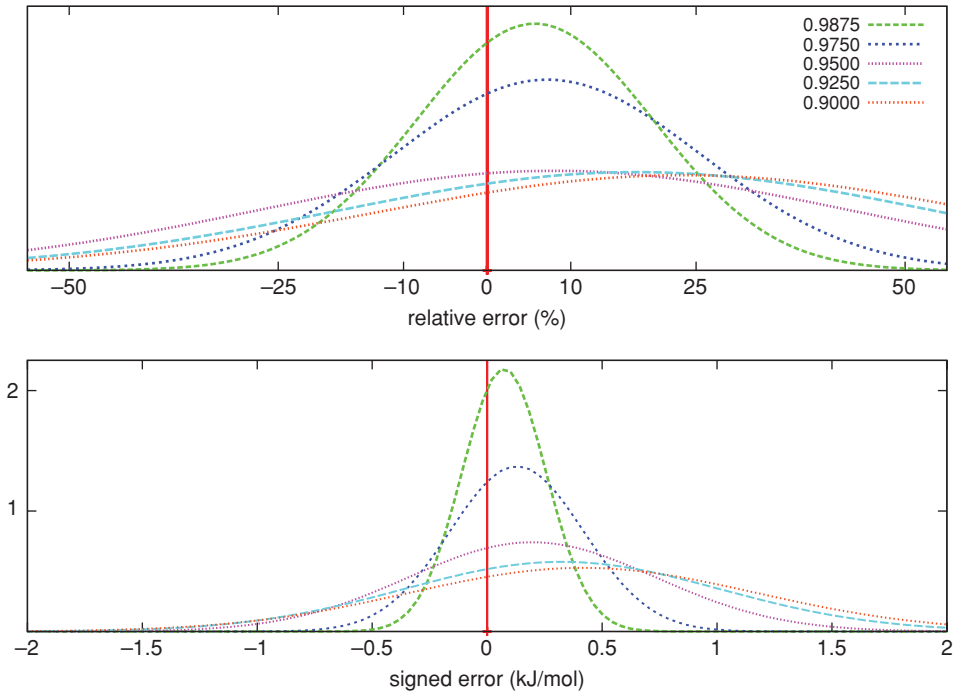

FIG. 9. Normal distribution of relative and signed errors for the T contribution to the total energy with the cc-pVTZ basis set (see text for details).

and nitrogen-oxide derivatives,[58] and we also evaluated the CCSDT and CCSDT(Q) total energies using the cc-pVTZ and cc-pVDZ basis sets, respectively. In this study, we selected 50 of these systems, namely, N_2, O_2, F_2, Cl_2, CN, OF, CCH, CHF, CHCl, HNO, H_2O_2, C_2H_2, CH_2F, CH_3F, CH_3Cl, CO_2, CF_2, CFCl, CCl_2, NO_2, FNO, ClNO, HOCO, HFCO, CHF_2, $CHCl_2$, CHClO, CHFCl, CH_2F_2, CH_2FCl, CH_2Cl_2, NH_2CO, CF_3, CF_2Cl, $CFCl_2$, CCl_3, CFClO, ClONO, F_2CO, CHF_3, $CHCl_3$, CHF_2Cl, $CHFCl_2$, CF_4, CF_2Cl_2, CF_3Cl, CCl_4, $CFCl_3$, $ClONO_2$, and C_6H_6 and calculated the CCSDT and CCSDT(Q) energies with the MP2 FNO approximation using various ε's for the truncation of the virtual space. Our results are analyzed based on the normal distribution of the errors with respect to the untruncated values calculated from the mean errors and standard deviations as described, e.g., in Ref. 59.

First, we investigated the distribution of the errors for the T contribution. For some molecules of the above test set, the T contribution is rather small, i.e., much less than 1 kJ/mol. As we have demonstrated in Sec. III B the MP2 FNO approach is able to estimate the order of magnitude of the T contribution even for molecules with such a small contribution; however, for these systems the relative error can be rather large, 100–200% in particular cases. In most of the applications, the T contribution for these systems can be neglected, but even if the MP2 FNO value of large relative error is used the absolute error is negligible. Thus, the molecules with a T contribution of smaller than 0.5 kJ/mol, that is, NO_2, HNO, HFCO, F_2, F_2CO, CF_4, C_2H_2, O_2, CHF_3, and H_2O_2, were excluded from the statistical analysis of relative errors. The average of the absolute values of T contributions for the restricted test set is about 1.9 kJ/mol. The normal distributions of the relative errors for the T contributions of the remaining 40 molecules can be found at the top of Fig. 9.

The distribution for $\varepsilon = 0.9875$ shows that 78% of the results have relative errors smaller than 20% and only 5% of the contributions have errors larger than 30%. The average of the retained virtual orbitals is 65% of the original virtual space, and thus the average speedup is roughly 11. Using $\varepsilon = 0.975$, 85% of the results have errors smaller than 30% and only 5% of the results have relative errors larger than 40%. The average of the retained virtual orbitals is 52% and the average speedup is 31. For $\varepsilon = 0.95$, the average of the retained virtual orbitals is 38% and the average speedup is close to 170. For most of the molecules, the latter truncation threshold can safely be applied except for benzene and the N_2 molecule for which the MP2 FNO approximation with $\varepsilon = 0.95$ completely fails. The T contribution for benzene is 1.57 kJ/mol, while the approximation gives 0.08 kJ/mol. Similarly, the respective results for the N_2 molecule are 0.74 kJ/mol and –0.1 kJ/mol. There are two further molecules in the restricted test set with large relative errors, FNO and CO_2, for which the absolute value of the T contribution is about 0.5 kJ/mol and the error is 0.3 kJ/mol, but the absolute magnitude of these errors are still below the 1 kJ/mol limit. The standard deviation with and without the above mentioned four molecules is 35% and 17%, respectively, i.e., these molecules are in charge of the significant part of the large standard deviation.

Beside the relative error, it can also be interesting to investigate the normal distribution of the absolute errors of the T contributions. The result of the analysis for the full 50-member test set can be found on the lower panel of Fig. 9. As the figure shows, the absolute errors for $\varepsilon = 0.9875$ are practically below the 0.5 kJ/mol limit, and for $\varepsilon = 0.975$ the errors still do not exceed 1 kJ/mol. As already discussed above, the application of $\varepsilon = 0.95$ might be dangerous. At this point, there are five molecules in the test set with T

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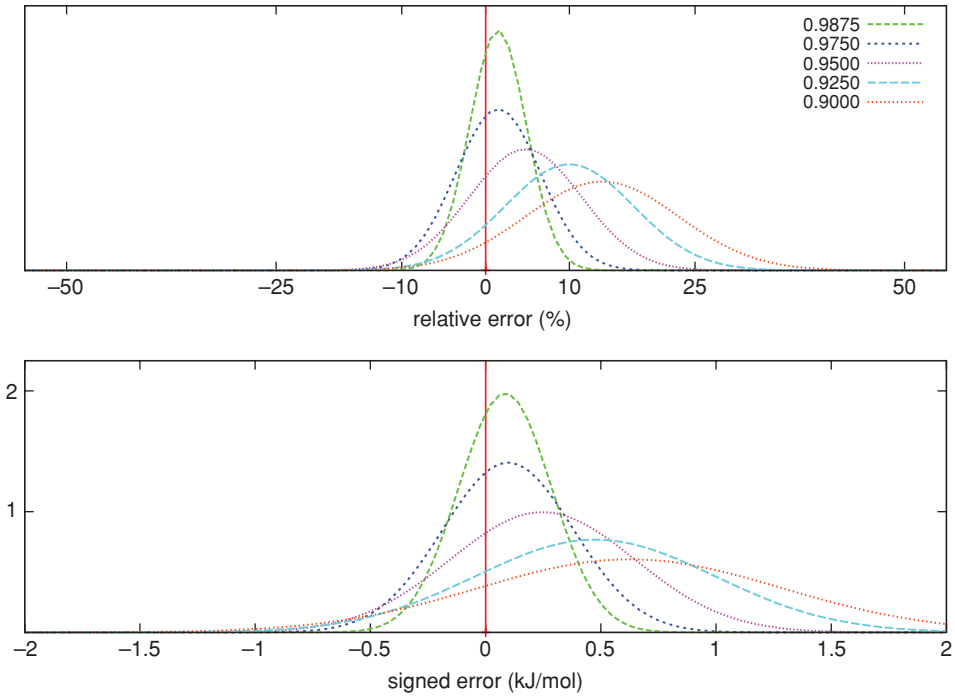

FIG. 10. Normal distribution of relative and signed errors for the (Q) contribution to the total energy with the cc-pVDZ basis (see text for details).

contributions of error larger than 1 kJ/mol. The benzene molecule was discussed above. The T contributions for the CCl_2, CCl_3, and CHCl molecules are roughly –3 kJ/mol, but the error of the approximation is about 1.2 kJ/mol. The fifth molecule with significant error is CN where the T contribution is large, –6 kJ/mol, and the approximation at $\varepsilon = 0.95$ only recovers –4.8 kJ/mol. As shown in Fig. 9, one can try to use smaller ε values to gain larger speedup, but the risk of significant error is larger.

For all the molecules of the test set the (Q) contributions are larger than 1 kJ/mol and their average is 4.2 kJ/mol. The normal distributions of the relative errors of the (Q) contributions for the 50-member test set are shown at the top of Fig. 10. Using the $\varepsilon = 0.9875$ threshold the relative errors are practically less than 10% for all the molecules. The ratio of the retained virtuals is 83%, which is substantially larger than the same value for the T calculations where larger, triple-ζ, basis set was applied. The average speedup is only 2.5. In the (Q) case, $\varepsilon = 0.975$ or $\varepsilon = 0.95$ offers a more balanced choice. For $\varepsilon = 0.975$, the results have relative errors smaller than 15%, while for $\varepsilon = 0.95$ only 5% of results exceeds the 15% barrier but 1% of results have error larger than 20%. The ratios of the retained virtual orbitals for $\varepsilon = 0.975$ and $\varepsilon = 0.95$ are 75 and 65%, and thus the average speedups are 5 and 11, respectively. At the $\varepsilon = 0.95$ value the largest relative error belongs to the CN molecule, where the (Q) contribution is –5.2 kJ/mol, while the approximation yields –4.2 kJ/mol.

The bottom of Fig. 10 shows the normal distribution of the absolute errors for the quadruples contribution. For $\varepsilon = 0.9875$ and $\varepsilon = 0.975$, the results are similar to those for the T correction. $\varepsilon = 0.9875$ practically keeps the error below 0.5 kJ/mol and for ε=0.975 the error is always smaller than 1 kJ/mol. There are four molecules where the application of $\varepsilon = 0.95$ results in error larger than 1 kJ/mol. The CN molecule has been mentioned above. The remaining three species $ClONO_2$, ClONO, and ClNO have relatively large (Q) contribution, –13.6, –10.3, and –8.2 kJ/mol, respectively. The error of the approximations at $\varepsilon = 0.95$ are 1.8, 1.2, and 1.3 kJ/mol, respectively. Again, one can take more risk to gain larger speedup and apply $\varepsilon = 0.925$. Here the mean value of the error is close to 0.5 kJ/mol and 16% of the molecules have error larger than 1 kJ/mol, but the speedup gained is a factor of 19. To accelerate the calculations for some of the molecules (such as CO_2 discussed in Sec. III B), the application of smaller ε is also possible but without *a priori* information (e.g., numerical studies on similar systems) this might be dangerous.

We should emphasize here that in the above analysis, to be conservative, we did not considered the size of the molecules. As we mentioned in Sec. II, the magnitude of the correlation contributions and probably the errors caused by the approximations grow with the size of the molecule. Thus, if the errors were divided, e.g., by the number of the chemical bonds, the absolute errors would be significantly smaller and would lie below 1 kJ/mol for the majority of the molecules even at lower values of ε.

As a conclusion, we have found that the relative errors for the results of the MP2 FNO approximation are substantially larger for the triples than for the quadruples contribution, but the distributions of the absolute errors for the various ε values are rather similar. For both the T and (Q) corrections the application of $\varepsilon = 0.975$ keeps the error below the 1 kJ/mol limit, but for most of the cases the $\varepsilon = 0.95$ gives also acceptable results, especially if the results are interpreted on a per bond basis. The speedups in the larger basis sets are obviously larger;

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TABLE I. Approximate T contributions to the total energy (kJ/mol) and the number of virtual orbitals (n_v) retained compared to the exact values for the cc-pVTZ and cc-pVQZ basis sets using $\varepsilon = 0.95$ and 0.975 values.

	cc-pVTZ						cc-pVQZ					
ε	0.95		0.975		1.0		0.95		0.975		1.0	
	T	n_v	T	n_v	T	n_v	T	n_v	T	n_v	T	n_v
CCH	–3.50	22	–3.61	30	–3.75	67	–3.30	26	–3.48	38	–3.33	133
CN	–4.76	17	–5.00	26	–6.02	53	–5.12	21	–5.12	32	–5.58	103
CHF	–1.10	22	–1.17	32	–1.70	66	–1.01	27	–1.28	40	–1.44	132
CH_2F	–0.95	29	–1.09	40	–1.24	79	–0.99	35	–1.17	47	–0.97	161
OF	–2.09	21	–2.25	30	–2.61	51	–2.14	27	–2.32	39	–2.52	101
CF_2	–0.80	28	–0.69	42	–1.05	78	–0.51	37	–0.69	54	–0.66	153
CHClO	–1.09	32	–1.25	45	–1.23	92	–1.35	39	–1.33	57	–0.84	183
ClNO	–1.41	28	–1.84	39	–1.90	78	–1.59	34	–1.74	51	–1.36	153
Cl_2	–1.29	25	–1.85	34	–2.14	51	–1.91	32	–2.27	44	–2.05	101
CCl_2	–2.17	33	–2.82	44	–3.30	78	–2.69	41	–3.22	68	–3.01	153

thus, the T contributions can be calculated either several tens or a hundred times faster, depending on the applied approximation but the typical speedup for the (Q) calculations, which is calculated in the smaller double-ζ basis set, is around 10.

For particular cases, the application of larger basis sets may be desired to improve the precision of the T or (Q) contribution or the size of the molecule can be larger than we have studied here. In these situations, the use of cost reduction techniques might be especially fruitful since we can hope that the larger number of basis functions contains larger amount of unimportant molecular orbitals which can be efficiently eliminated by the MP2 FNO method. To check this idea, we have performed a series of calculations for ten molecules of the above test set (see Tables I and II) with the cc-pVQZ basis set for the T and cc-pVTZ for the (Q) calculations. The results are compiled and compared to the smaller basis set results in Tables I and II. To investigate the effect of the system size on the number of retained virtual orbitals, we have also studied the linear C_nH_{2n+2} alkane chains up to $n = 7$ at standardized geometries with tetrahedral bond angles and C–C and C–H bond distances of 1.55 and 1.09 Å, respectively (see Fig. 11).

As one can see, the application of larger basis sets significantly improves the stability of the approximate T contribution. The absolute errors of the approximate values in the cc-pVQZ basis are significantly lower for the CN and CHF molecules if $\varepsilon = 0.975$ is used, and for the CN, CHF, ClNO, Cl_2, and CCl_2 molecules if $\varepsilon = 0.95$ is applied. The only molecule where the absolute errors are consistently larger in the cc-pVQZ basis is CHClO. Note, that with the application of the larger basis set the number of retained virtual orbitals is slightly larger at a given ε, but the ratio of the number retained and original virtual orbitals is substantially smaller.

In the (Q) case, the stability of the results for the test set is not improved by the application of the larger, cc-pVTZ basis set. The absolute errors of the approximations are similar or slightly larger for the larger basis sets. This later is especially true for the CN, ClNO, and CCl_2 molecules, where the use of larger basis set significantly modifies the (Q) contributions. As we expect, increasing the number of basis functions by the application of the larger cc-pVTZ basis significantly reduces the fraction of the retained virtual orbitals.

One arrives at different conclusions if the number of basis functions increases due to the increasing system size. In Fig. 11, the ratio of the retained and the original number of virtual orbitals is plotted against the system size for various values of ε. As the figure shows the percentage of the retained virtual orbitals practically does not change from $n = 2$ with both basis sets. This suggest that the number of the dropped virtual orbitals for a group of related linear molecules, and thus the theoretical speedup is rather independent of the

TABLE II. Approximate (Q) contributions to the total energy (kJ/mol) and the number of virtual orbitals (n_v) retained compared to the exact values for the cc-pVDZ and cc-pVTZ basis sets using $\varepsilon = 0.95$ and 0.975 values.

	cc-pVDZ						cc-pVTZ					
ε	0.95		0.975		1.0		0.95		0.975		1.0	
	(Q)	n_v	(Q)	n_v	(Q)	n_v	(Q)	n_v	(Q)	n_v	(Q)	n_v
CCH	–2.25	15	–2.28	17	–2.57	26	–2.23	22	–2.46	30	–3.12	67
CN	–4.17	12	–4.64	16	–5.22	21	–4.25	17	–4.59	26	–6.09	53
CHF	–1.66	15	–1.69	18	–1.80	31	–1.82	22	–1.35	42	–1.63	66
CH_2F	–1.46	18	–1.45	20	–1.40	29	–1.16	29	–1.01	40	–1.06	79
OF	–2.86	15	–3.06	17	–3.16	19	–3.00	21	–2.81	30	–2.98	51
CF_2	–2.87	20	–3.17	23	–3.25	30	–3.40	28	–2.48	42	–2.78	78
CHClO	–3.85	21	–4.04	24	–3.96	44	–4.27	32	–3.88	44	–4.38	101
ClNO	–6.84	20	–7.39	30	–8.16	23	–7.95	28	–8.15	39	–9.41	78
Cl_2	–1.62	16	–1.78	19	–1.86	17	–2.53	25	–2.52	34	–2.58	49
CCl_2	–2.81	21	–2.85	30	–3.23	24	–3.88	33	–4.05	44	–4.79	78

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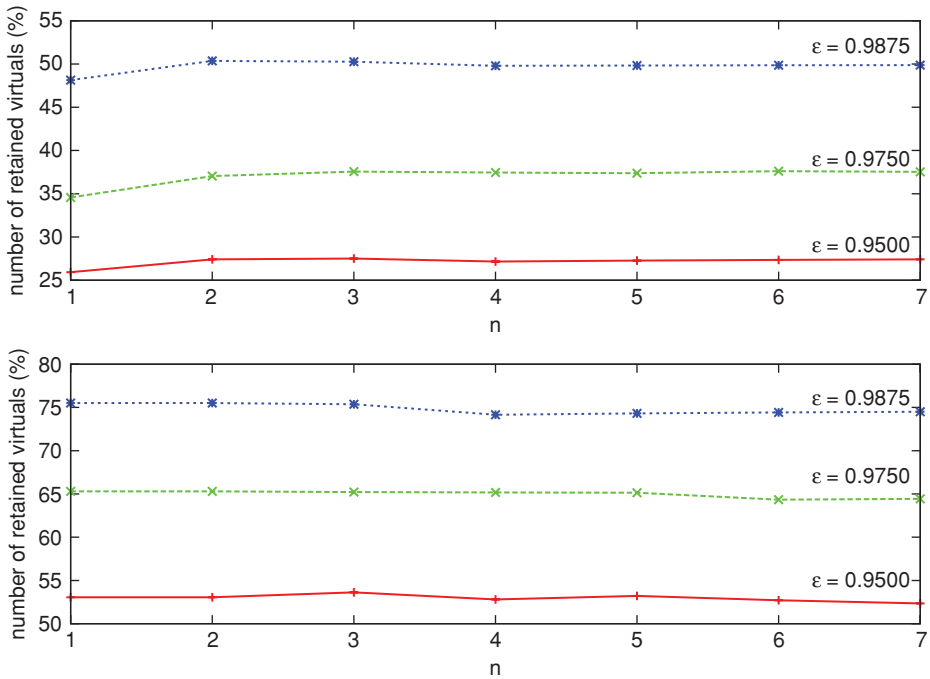

FIG. 11. The number of retained virtual orbitals for C_nH_{2n+2} alkane chains as a function of the system size with the cc-pVTZ (top) and cc-pVDZ (bottom) basis sets (see text for details).

system size. We should mention, however, that the measured speedups for larger molecules are expected to be significantly larger than the theoretical ones for the reasons outlined in Sec. III E.

D. Correlation contributions to thermochemical properties

The primary purpose of this study is to develop reduced-cost high-order CC methods for the calculation of the iterative triples and quadruples corrections used in high-precision model chemistries. Since the latter are most frequently used for the calculation of thermochemical properties, we tested the performance of the MP2 FNO approach for reaction heats. The calculation of energy differences, such as reaction heats, poses some challenge for the approximate schemes since the errors might be additive. The starting point of our analysis is the above test set. We calculated the iterative triples and quadruples contributions to the reaction heats of the 56 chemical reactions collected in Table III using the MP2 FNO scheme and employing various thresholds for the truncation of the virtual space. The convergence of the approximate T and (Q) corrections with the level of truncation is demonstrated in Fig. 12.

The average of the absolute value of the T contributions for the 56-member test set is 1.3 kJ/mol. For a significant part of the test reactions the T contribution to the reaction heat is relatively small, i.e., less than 1 kJ/mol. Even in those cases the results are frequently obtained as a difference of larger contributions in the order of 2–3 kJ/mol.

To investigate the effect of the error of T contributions for individual molecules on the reaction heats, we can compare the normal distributions of the molecular errors and the errors of reaction heats. For $\varepsilon = 0.9875$, the standard deviation is 0.25 kJ/mol, which is slightly larger than the standard deviation for the molecular contributions, 0.18 kJ/mol, at the same level of approximation. As the upper panel of Fig. 12 shows, the mean signed error is very close to zero, and the error of the reaction heats is less than 0.5 kJ/mol for most of the cases. In the test set, only reactions 42 and 47 of Table III have slightly larger error. The T contribution for these reactions are 6.8 and –3.6 kJ/mol and the approximation yields 6.3 and –2.9 kJ/mol, respectively.

If $\varepsilon = 0.975$ is applied the standard deviations for the molecular contributions and reaction heats are 0.29 and 0.36 kJ/mol, respectively. The center of the distributions of the molecular errors and the errors of reactions are located at 0.13 and 0.03 kJ/mol, respectively, i.e., the mean signed error for reactions is closer to zero. Consequently, even if the distribution of the errors for the reaction energies are wider compared to the molecular errors, the errors are kept in the same range since the larger standard deviation for the reaction heats is compensated by the shift of the mean signed error toward zero. We can say again that for $\varepsilon = 0.975$, the errors for almost all the reactions lie in the -1 to 1 kJ/mol interval. The only exception is reaction 42 again, where the MP2 FNO approximation yields 5.6 kJ/mol.

If $\varepsilon = 0.95$ is applied the standard deviation is larger than it was for the molecular contributions but as a consequence of the error compensation discussed above the quality of the approximation is similar. In the test set, there are six reactions where the error at $\varepsilon = 0.95$ is larger than 1 kJ/mol. These reactions have relatively large T contributions. For reactions 42 and 47 mentioned above, the approximation give 4.8 and –2.0 kJ/mol, respectively. Further, for reactions 36, 43, 53,

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TABLE III. Test set for reaction heat calculations.

1	$CH_3F \longrightarrow CHF + H_2$	29	$\frac{1}{2}N_2 + O_2 \longrightarrow NO_2$
2	$CH_2FCl \longrightarrow CFCl + H_2$	30	$F_2CO + \frac{1}{2}O_2 \longrightarrow F_2 + CO_2$
3	$CFClO + ClNO \longrightarrow CFCl + ClONO$	31	$HOCO \longrightarrow O_2 + \frac{1}{2}CCH + \frac{1}{4}H_2$
4	$ClONO + CHF \longrightarrow CHFCl + NO_2$	32	$\frac{1}{2}CO_2 + \frac{1}{2}CHF_3 \longrightarrow HFCO + \frac{1}{2}F_2CO$
5	$CH_3Cl + FNO \longrightarrow ClNO + CH_3F$	33	$CHF_2Cl + \frac{1}{2}O_2 \longrightarrow HFCO + \frac{1}{2}F_2 + \frac{1}{2}Cl_2$
6	$CHF_2 + CF_2 \longrightarrow CF_3 + CHF$	34	$CF_3 + CHFCl_2 \longrightarrow CF4 + CHCl_2$
7	$FNO + CHF_2Cl \longrightarrow ClNO + CHF_3$	35	$CH_2F + NO_2 \longrightarrow HNO + HFCO$
8	$CH_2FCl + CO_2 \longrightarrow HFCO + CHClO$	36	$CHF + HOCO \longrightarrow CH_2F + CO_2$
9	$CHFCl \longrightarrow CFCl + \frac{1}{2}H_2$	37	$CF_2Cl_2 \longrightarrow CCl_2 + F_2$
10	$CF_3Cl + NO_2 \longrightarrow ClONO + CF_3$	38	$CHCl_2 + CFCl_2 \longrightarrow CCl_4 + CHF$
11	$CF_2 + CF_2Cl_2 \longrightarrow CF4 + CCl_2$	39	$CHCl_3 + NO_2 \longrightarrow ClONO + CHCl_2$
12	$CF_2 + CF_3Cl \longrightarrow CF_2Cl + CF_3$	40	$HOCO \longrightarrow CO_2 + \frac{1}{2}H_2$
13	$CCl_3 + CHF_2 \longrightarrow CHCl_3 + CF_2$	41	$CHF_2 \longrightarrow F_2 + \frac{1}{2}C_2H_2$
14	$ClONO \longrightarrow \frac{1}{2}Cl_2 + NO_2$	42	$CN + O_2 \longrightarrow CO_2 + \frac{1}{2}N_2$
15	$F_2CO + CH_3F \longrightarrow CH_2F_2 + HFCO$	43	$ClONO_2 + \frac{1}{2}N_2 \longrightarrow ClNO + NO_2$
16	$CHFCl_2 + F_2CO \longrightarrow CHClO + CF_3Cl$	44	$\frac{1}{2}O_2 + \frac{1}{2}F_2 \longrightarrow OF$
17	$CH_2F_2 + F_2CO \longrightarrow HFCO + CHF_3$	45	$CHClO + \frac{1}{2}O_2 \longrightarrow HOCO + \frac{1}{2}Cl_2$
18	$CFCl_2 + CHF_2Cl \longrightarrow CCl_3 + CHF_3$	46	$CF_4 \longrightarrow CF_2 + F_2$
19	$CHFCl + \frac{1}{2}H_2 \longrightarrow CH_2FCl$	47	$\frac{1}{2}C_2H_2 + \frac{1}{2}CCl_4 \longrightarrow CHCl + \frac{1}{2}CCl_2$
20	$CH_2Cl_2 + CHF_3 \longrightarrow CH_2FCl + CHF_2Cl$	48	$\frac{1}{3}C6H6 \longrightarrow CCH + \frac{1}{2}H_2$
21	$\frac{1}{2}CCl_2 + HFCO \longrightarrow CHClO + \frac{1}{2}CF_2$	49	$ClONO_2 \longrightarrow ClONO + \frac{1}{2}O_2$
22	$ClNO + \frac{1}{2}O_2 \longrightarrow NO_2 + \frac{1}{2}Cl_2$	50	$CCH + \frac{1}{2}H_2 \longrightarrow C_2H_2$
23	$\frac{1}{2}CCl_4 + \frac{1}{2}CH_2F_2 \longrightarrow CHFCl_2$	51	$\frac{1}{2}C_2H_2 + OF \longrightarrow HFCO$
24	$CF_2Cl + \frac{1}{2}O_2 \longrightarrow F_2CO + \frac{1}{2}Cl_2$	52	$C_2H_2 \longrightarrow \frac{1}{3}C_6H_6$
25	$\frac{1}{3}CH_2F + \frac{1}{3}CHFCl + \frac{1}{3}CFCl_2 \longrightarrow CHFCl$	53	$CHCl + \frac{1}{2}O_2 \longrightarrow CHClO$
26	$\frac{2}{3}NO_2 + \frac{2}{3}HOCO + \frac{1}{3}H_2 \longrightarrow O_2 + \frac{2}{3}NH_2CO$	54	$CFCl_3 + CHF \longrightarrow CFCl_2 + CHFCl$
27	$\frac{1}{2}CO_2 + \frac{1}{2}CH_2Cl_2 \longrightarrow CHClO$	55	$CCl_4 \longrightarrow Cl_2 + CCl_2$
28	$HNO \longrightarrow \frac{1}{2}H_2O_2 + \frac{1}{2}N_2$	56	$CH_3Cl \longrightarrow CHCl + H_2$

and 55 the absolute magnitude of the T contributions are underestimated by a value slightly larger than 1 kJ/mol.

The mean of the absolute value of the (Q) contributions for the test set in Table III is 1.1 kJ/mol, which is close to the value we have found for the T contribution. In the test set, the typical (Q) contributions to the reaction heats are calculated as a difference of quantities in the 5–9 kJ/mol interval.

As one can see in Fig. 12, the error compensation that we have described for the triples contribution to reaction heats works also for the quadruples contributions. The distributions

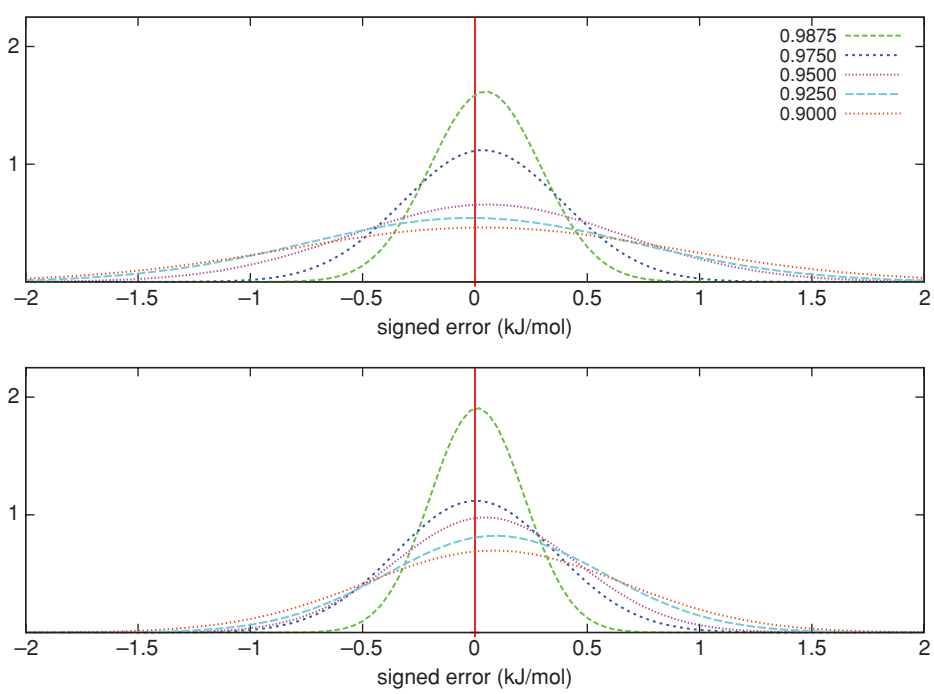

FIG. 12. Normal distribution of errors of the T (top) and (Q) (bottom) contributions to the heats of formation (see text for details).

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at the various levels of approximation are slightly wider for the contributions to reaction heats than those for individual molecules, but the centers of the error distributions are closer to zero in the case of reaction heats. As a consequence, the absolute errors for the reactions obtained at various levels of approximation do not exceed the absolute errors of the molecules. If $\varepsilon = 0.9875$ is applied, the error in the reaction heats will be smaller than 0.5 kJ/mol for most of the reactions. There are three exceptions in the test set, reactions 39, 42, and 49, where the errors are larger than 0.5 kJ/mol. For these reactions the approximation gives –1.6, 1.9, and 0.2 kJ/mol while the original values are –2.1, 2.6, and 0.8 kJ/mol, respectively. If $\varepsilon = 0.975$ or $\varepsilon = 0.95$ are used, the error is still smaller than 1 kJ/mol for most of the cases. In our test set, only reaction 29 has error larger than 1 kJ/mol, where the approximation, e.g., at $\varepsilon = 0.95$ yields –0.7 kJ/mol instead of –1.7 kJ/mol. We can also observe that the error compensation for $\varepsilon = 0.925$ or 0.90 still works surprisingly well. Even for these parameters most of the absolute errors for reaction heats lie below the 1 kJ/mol limit, and the maximum absolute error is lower than 1.5 kJ/mol.

As a conclusion, we can say that the magnitude of the absolute errors for the heats of reaction and that for molecules are in the same range for all the investigated ε values. Similarly to the molecular contributions, the performance of the approximations for reaction heats is better for the quadruples than for the triples contributions.

E. Timings

In accordance with the above results about 50 and 60% of the virtual orbitals can be dropped at the CCSDT and CCSDT(Q) calculations for butadiene and benzene, respectively, using the orbital transformation approaches, and thus, in principle, speedups of 32 and about 100 can be achieved. One may be interested in the effective speedups which can be measured using our implementation. Therefore we compiled computation times in Table IV for the butadiene and benzene molecules, which are large enough for such analysis. The calculations were carried out on machines with Intel Xeon E3110 3.00 GHz CPUs and 8 Gb main memory.

The effective speedups in the smaller basis set fall behind the theoretical ones especially for the CCSDT method. In contrast, the measured speedup for the CCSDT methods with the cc-pVTZ basis set is very close to the theoretical one for butadiene and much larger than the theoretical value for benzene. The latter unexpected speedup is a consequence of the considerably larger memory requirement of the untruncated calculation. The size of the cluster amplitude matrices, integral lists, and intermediates scales as the third or fourth power of the virtual-space size in the case of a CCSDT calculation, thus only a smaller portion of the elements of these arrays can fit into the memory at the same time for the untruncated calculation, which results in increased I/O demand. We think that due to the steep memory requirement of the calculations the ratio of the measured and theoretical speedups is even higher for bigger systems.

F. Applications to larger molecules

To show the potential of the cost reduction techniques studied, we have performed large scale CCSDT and CCSDT(Q) calculations for the naphthalene and anthracene molecules. For naphthalene, both the iterative triples and the quadruples contributions have been determined using the cc-pVTZ and cc-pVDZ basis sets, respectively, while for anthracene we were only able to compute the quadruples correction, which have been evaluated with the cc-pVDZ basis set. The energy calculations were performed at geometries optimized by the CCSD(T) and MP2 methods, respectively, and the cc-pVTZ basis set. We employed the MP2 FNO approach and chose the size of the virtual space so that the calculations can be carried out within reasonable time. For naphthalene 48 electrons of the 18 atoms were correlated as well as 96 and 175 optimized MOs were retained in the double- and triple-ζ sets, respectively. For anthracene, a 24-atom molecule, 66 electrons were correlated using 112 optimized active orbitals. These truncations correspond to ε's of 0.95, 0.985, and 0.916, respectively.

The calculations were carried out on a machine with two quad-core Intel Xeon (5410) 2.33 GHz CPUs and used maximum of 4.8, 7.2, and 18.5 Gb memory, respectively. The results of the calculations and the required computational times are collected in Tables V and VI, where we also give estimates for the computation times of the calculations carried out with the untruncated basis set. The latter estimates are simply calculated on the basis of the theoretical speedup, $(n_v/n_{av})^4$ for CCSD(T) and $(n_v/n_{av})^5$ for CCSDT and CCSDT(Q). In fact, the untruncated calculations are currently not feasible due to their enormous memory and disk-space requirement.

As we saw in the previous benchmark calculations, for naphthalene the above ε thresholds are enough to keep the

TABLE IV. Timings (hours) for the full and the truncated calculations as well as theoretical and measured speedups for the butadiene and benzene molecules using the CCSDT and CCSDT(Q) methods.

			No. of virtuals		CPU time		Speedup	
Molecule	Basis	Method	Full	Truncated	Full	Truncated	Theoretical	Measured
Butadiene	cc-pVDZ	CCSDT	71	28	2.66	0.05	104.8	58.4
		(Q) correction	71	28	13.17	0.18	104.8	74.7
	cc-pVTZ	CCSDT	189	94	229.29	7.55	32.9	30.4
Benzene	cc-pVDZ	CCSDT	93	37	8.05	0.26	100.3	30.4
		(Q) correction	93	37	34.99	0.54	100.3	64.5
	cc-pVTZ	CCSDT	243	122	915.99	10.84	31.3	84.5

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TABLE V. Total energies (E_h) and timings (day) for the naphthalene molecule with the cc-pVDZ and cc-pVTZ basis sets using the MP2 FNO approach. $\varepsilon = 0.95$ and $\varepsilon = 0.985$ were applied for the (Q) and T contributions, respectively. Estimated timings for the untruncated calculations are given in the last column.

Basis	Method	Energy	CPU time	Estimated CPU time
cc-pVTZ	CCSD(T)	–385.020008	0.03	1.18
	CCSDT	–385.019000	17.56	1726.23
	T-(T)	0.001008		
cc-pVDZ	CCSDT	–384.581706	1.35	46.28
	CCSDT(Q)	–384.586393	6.94	237.94
	(Q)	–0.004687	5.59	191.66

error coming from the truncation of the virtual space below the desired limit, while the accuracy of the anthracene results merits some discussion. The approximate results for naphthalene and anthracene can be compared to those obtained for the physically similar benzene molecule where the exact CCSDT and CCSDT(Q) results are also available. It is important to note that for the benzene molecule the relative error of the (Q) contribution at $\varepsilon = 0.915$ is 12%, and the (Q) contribution is practically a linear function of ε in the $\varepsilon = 0.9$–1.0 range. We expect a similar behavior for the naphthalene and anthracene molecules. To check this, we have performed four calculations for the (Q) contribution for naphthalene in the $\varepsilon = 0.9$–0.95 range. The (Q) contributions for the $\varepsilon = 0.9$, 0.915, 0.925, and 0.95 values are –4.296, –4.399, –4.495, and –4.687 mE_h, respectively. This shows a linear dependence, and the contributions can be extrapolated to $\varepsilon = 1.0$, which gives –5.082 mE_h. Thus, the relative error of the (Q) contribution evaluated at $\varepsilon = 0.915$ is about 13%, which is very close that of benzene. Consequently, we expect that the dependence of the error on the truncation of the virtual space is also similar for anthracene, and the relative error for the calculated (Q) contribution does not exceed 15%.

The results clearly demonstrate the strong influence of higher-order correlation contributions on the total energies for these molecules and also show that the developed reduced-cost methods allow for the evaluation of the high-order correlation contributions within reasonable time. The results also indicate that the evaluation of the quadruples correction is less expensive than that for the iterative triples, at least for the currently feasible systems. Though we can now calculate the effect of quadruple excitations for as large molecules as anthracene, for high-accuracy calculations we would also need the corresponding triples correction, which is currently available for molecules of naphthalene size. Thus, all in all, the scope of high-accuracy calculations can be extended to the systems of this size.

TABLE VI. Total energies (E_h) and timings (day) for the anthracene molecule with the cc-pVDZ basis set using the MP2 FNO approach ($\varepsilon = 0.916$). Estimated timings for the untruncated calculations are given in the last column.

Method	Energy	CPU time	Estimated CPU time
CCSDT	–537.548073	3.5	355.0
CCSDT(Q)	–537.554784	11.6	1176.5
(Q)	–0.006711	8.1	821.5

IV. CONCLUDING REMARKS

We have studied the AS CC, MP2 FNO, and OVOs cost-reduction techniques for the approximation of iterative triples as well as quadruples contributions for the purpose of high-precision calculations. It has been shown that the AS CC approach can be used to reduce the computational price, but the gain in the speed is significantly lower than that for the MP2 FNO or OVOs methods. No remarkable difference has been observed between the performance of the MP2 FNO and OVOs methods, though the simplicity of the MP2 FNO approach with respect to the latter makes it more favorable.

Our MP2 FNO test calculations demonstrate that the use of the $\varepsilon = 0.95$ threshold for the cumulative population of the retained virtual orbitals is enough to obtain the quadruples contribution with a reasonable loss in accuracy thereby reducing the computational time by about an order of magnitude. For the approximation of the triples contribution a larger ε value, e.g., 0.975, might be needed. Despite this the achievable speedup, a factor of several tens, can be larger than that for the quadruples since these calculations are performed with a larger, triple-ζ basis set. With even larger basis sets the speedup can be significantly better. We have also shown that the errors of the heats of formation calculated by the reduced-cost CC methods are in the same range as the errors of the individual molecules. The calculations for naphthalene and anthracene demonstrate that the MP2 FNO method allows for the calculation of the iterative triples and quadruples contributions for molecules of more than 40 electrons on commodity computers.

We note that we also explored several further ways for the reduction of the costs of high-order CC methods including the SVD of the cluster amplitude matrices for both iterative and perturbative CC models, the Cholesky decomposition of the energy denominators in the case of CCSDT(Q), or the factorization of the two-electron integral list. Though a moderate reduction in the computational expenses can also be achieved using these techniques, they cannot compete with the orbital transformation approaches discussed here. In a subsequent paper, we will show that the combination of the MP2 FNO scheme with localized approaches results in an even more robust cost-reduction technique for high-order CC methods and allows for high-accuracy calculations for significantly larger molecules.

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APPENDIX: WORKING EQUATIONS DERIVED FROM THE OVERLAP FUNCTIONAL

Our approach follows the guidelines of Neogrády *et al.* and the derivation of the working equations connected to the CC energy functional.[37]

To find the maximum of L_{overlap} as a function of the rotational parameters $U_{a'a}$ with keeping the orthogonality of MOs the technique of Lagrange multipliers is invoked, and a modified functional is defined as

$$\widetilde{L} = L_{\text{overlap}} + \sum_{a^*b^*} \lambda_{a^*b^*}(\delta_{a^*b^*} - \langle a^*|b^*\rangle) = \sum_i \sum_{a^*} t_i^{a^*} t_i^{a^*} + \frac{1}{4}\sum_{ij}\sum_{a^*b^*} t_{ij}^{a^*b^*} t_{ij}^{a^*b^*} + \sum_{a^*b^*} \lambda_{a^*b^*}(\delta_{a^*b^*} - \langle a^*|b^*\rangle), \tag{8}$$

with $\lambda_{a^*b^*}$ as the multipliers. Note that as a consequence of the restricted summation for the virtual indices of $\hat{T}^*_{\text{CCSD}}$, the virtual indices in Eq. (8) are also restricted to the active orbitals. To explicitly introduce the rotational parameters, Eq. (6) is substituted into Eq. (8) to obtain

$$\widetilde{L} = \sum_i \sum_{a^*} \sum_{pq} U_{a^*p} U_{a^*q} t_i^p t_i^q + \frac{1}{4}\sum_{ij}\sum_{a^*b^*}\sum_{pqmn} U_{a^*p} U_{b^*q} t_{ij}^{pq} U_{a^*m} U_{b^*n} t_{ij}^{mn} + \sum_{a^*b^*} \lambda_{a^*b^*}\left(\delta_{a^*b^*} - \sum_p U_{a^*p} U_{b^*p}\right). \tag{9}$$

The condition of stationarity of $\widetilde{L}$ with respect to the elements of $\mathbf{U}$ results in the

$$\frac{\partial \widetilde{L}}{\partial U_{c^*s}} = 2\sum_i \sum_q U_{c^*q} t_i^s t_i^q + \sum_{ij}\sum_{b^*}\sum_{qmn} U_{b^*q} t_{ij}^{sq} U_{c^*m} U_{b^*n} t_{ij}^{mn} - 2\sum_{b^*} \lambda_{c^*b^*} U_{b^*s} = 0 \tag{10}$$

equations for the unknown parameters. Rearranging Eq. (10) leads to a self-consistent equation for matrix $\mathbf{U}$,

$$\mathbf{Q}(\mathbf{U})\mathbf{U}^{\mathbf{T}} = \mathbf{U}^{\mathbf{T}}\lambda^{\mathbf{T}}, \tag{11}$$

where

$$Q_{sq}(\mathbf{U}) = \sum_i t_i^s t_i^q + \frac{1}{2}\sum_{ij}\sum_{mn} t_{ij}^{qm} t_{ij}^{sn} P_{mn}(\mathbf{U}), \tag{12}$$

and

$$P_{mn}(\mathbf{U}) = \sum_{b^*} U_{b^*m} U_{b^*n}. \tag{13}$$

To solve Eq. (11), one can utilize the $\mathbf{U}\mathbf{U}^{\mathbf{T}} = \mathbf{1}$ relation and rewrite the equation as

$$\mathbf{U}\mathbf{Q}(\mathbf{U})\mathbf{U}^{\mathbf{T}} = \lambda^{\mathbf{T}}, \tag{14}$$

where matrix λ is symmetric. The subspace of the optimized virtual orbitals is obviously invariant to the rotation of the active orbitals; thus, we can suppose that matrix λ is diagonal. To find the physical interpretation of the λ_{a^*} eigenvalues of matrix λ, let us transform the cluster amplitudes to the optimized basis set as

$$t_{ij}^{\prime a^*b^*} = \sum_{cd} U_{a^*c} U_{b^*d} t_{ij}^{cd}. \tag{15}$$

Utilizing that the t_i^a amplitudes are considered zero at first order, and using Eq. (8) as well as Eqs. (12)–(15) one can find a relation between the trace of matrix λ and functional L_{overlap} which reads as

$$L_{\text{overlap}} = \frac{1}{4}\sum_{ij}\sum_{a^*b^*} t_{ij}^{\prime a^*b^*} t_{ij}^{\prime a^*b^*} = \frac{1}{2}\sum_{a^*} \lambda_{a^*}, \tag{16}$$

where the summation is restricted to the active orbitals. This relation shows that λ_{a^*} is proportional to the contribution of virtual orbital a^* to the overlap.

Equation (14) is a recursive equation for matrix $\mathbf{U}$ which can be solved in an iterative manner similar to the process used in the HF self-consistent field problem. In practice, at each iteration step matrix λ is calculated for the whole virtual space, and to maximize Eq. (16) those orbitals are chosen as active which have large λ_{a^*} values. Finally, matrix $\mathbf{P}$ is calculated with the selected active orbitals according to Eq. (13).

[1] J. M. L. Martin and G. de Oliveira, J. Chem. Phys. **111**, 1843 (1999).
[2] T. A. Ruden, T. Helgaker, P. Jørgensen, and J. Olsen, Chem. Phys. Lett. **371**, 62 (2003).
[3] A. D. Boese, M. Oren, O. Atasoylu, J. M.L. Martin, M. Kállay, and J. Gauss, J. Chem. Phys. **120**, 4129 (2004).
[4] A. Tajti, P. G. Szalay, A. G. Császár, M. Kállay, J. Gauss, E. F. Valeev, B. A. Flowers, J. Vázquez, and J. F. Stanton, J. Chem. Phys. **121**, 11599 (2004).
[5] T. Helgaker, T. A. Ruden, P. Jørgensen, J. Olsen, and W. Klopper, J. Phys. Org. Chem. **17**, 913 (2004).
[6] Y. J. Bomble, J. Vazquéz, M. Kállay, C. Michauk, P. G. Szalay, A. G. Császár, J. Gauss, and J. F. Stanton, J. Chem. Phys. **125**, 064108 (2006).
[7] M. E. Harding, J. Vázquez, B. Ruscic, A. K. Wilson, J. Gauss, and J. F. Stanton, J. Chem. Phys. **128**, 114111 (2008).
[8] A. Karton, E. Rabinovich, J. M.L. Martin, and B. Ruscic, J. Chem. Phys. **125**, 144108 (2006).
[9] D. Feller, K. A. Peterson, and D. A. Dixon, J. Chem. Phys. **129**, 204105 (2008).
[10] K. Raghavachari, G. W. Trucks, J. A. Pople, and M. Head-Gordon, Chem. Phys. Lett. **157**, 479 (1989).
[11] J. Noga and R. J. Bartlett, J. Chem. Phys. **86**, 7041 (1987); J. Noga and R. J. Bartlett, J. Chem. Phys. **89**, 3401 (1989), Erratum.
[12] Y. J. Bomble, J. F. Stanton, M. Kállay, and J. Gauss, J. Chem. Phys. **123**, 054101 (2005).
[13] M. Kállay and J. Gauss, J. Chem. Phys. **123**, 214105 (2005).
[14] We note that in this study we use the CCSDT(Q) method as defined in Ref. 12, and this definition is different from that given in Ref. 60.
[15] N. Oliphant and L. Adamowicz, J. Chem. Phys. **94**, 1229 (1991).
[16] S. A. Kucharski and R. J. Bartlett, J. Chem. Phys. **97**, 4282 (1992).
[17] M. Kállay and P. R. Surján, J. Chem. Phys. **115**, 2945 (2001).
[18] P. Piecuch, N. Oliphant, and L. Adamowicz, J. Chem. Phys. **99**, 1875 (1993).
[19] P. Piecuch, S. A. Kucharski, and R. J. Bartlett, J. Chem. Phys. **110**, 6103 (1999).
[20] M. Kállay, P. G. Szalay, and P. R. Surján, J. Chem. Phys. **117**, 980 (2002).

25 January 2026 17:33:09

[21] A. Köhn and J. Olsen, J. Chem. Phys. **125**, 174110 (2006).
[22] P. D. Fan and S. Hirata, J. Chem. Phys. **124**, 104108 (2006).
[23] K. Kowalski and P. Piecuch, J. Chem. Phys. **115**, 643 (2001).
[24] K. Kowalski and P. Piecuch, Chem. Phys. Lett. **347**, 237 (2001).
[25] K. Kowalski, S. Hirata, M. Włoch, P. Piecuch, and T. L. Windus, J. Chem. Phys. **123**, 074320 (2005).
[26] P.-D. Fan, M. Kamiya, and S. Hirata, J. Chem. Theor. Comp. **3**, 1036 (2007).
[27] W. Meyer, J. Chem. Phys. **58**, 1017 (1973).
[28] P. J. Hay, J. Chem. Phys. **59**, 2468 (1973).
[29] R. Ahlrichs, H. Lischka, V. Staemmler, and W. Kutzelnigg, J. Chem. Phys. **62**, 1225 (1975).
[30] C. Edmiston and M. Krauss, J. Chem. Phys. **45**, 1833 (1966).
[31] T. R. Barr and E. R. Davidson, Phys. Rev. A **1**, 644 (1970).
[32] C. Sosa, J. Geersten, G. W. Trucks, R. J. Bartlett, and J. A. Franz, Chem. Phys. Lett. **159**, 148 (1989).
[33] A. Landau, K. Khistyaev, S. Dolgikh, and A. I. Krylov, J. Chem. Phys. **132**, 014109 (2010).
[34] L. Adamowicz and R. J. Bartlett, J. Chem. Phys. **86**, 6314 (1987).
[35] L. Adamowicz, R. J. Bartlett, and J. Sadlej, J. Chem. Phys. **88**, 5749 (1988).
[36] M. Urban, J. Noga, S. J. Cole, and R. J. Bartlett, J. Chem. Phys. **83**, 4041 (1985).
[37] P. Neogrády, M. Pitoňák, and M. Urban, Mol. Phys. **103**, 2141 (2005).
[38] A. G. Taube and R. J. Bartlett, Collect. Czech. Chem. Commun. **70**, 837 (2005).
[39] M. Pitoňák, F. Holka, P. Neogrády, and M. Urban, J. Mol. Struct. (THEOCHEM) **768**, 79 (2006).
[40] W. Klopper, J. Noga, H. Koch, and T. Helgaker, Theor. Chem. Acc. **97**, 164 (1996).
[41] N. H.F. Beebe and J. Linderberg, Int. J. Quantum Chem. **7**, 683 (1977).
[42] H. Koch, A. Sánchez de Merás, and T. B. Pedersen, J. Chem. Phys. **118**, 9481 (2003).
[43] T. B. Pedersen, A. M. Sánchez de Merás, and H. Koch, J. Chem. Phys. **120**, 8887 (2004).
[44] H. Koch and A. M. Sánchez de Merás, J. Chem. Phys. **113**, 508 (2000).
[45] T. Kinoshita, O. Hino, and R. J. Bartlett, J. Chem. Phys. **119**, 7756 (2003).
[46] O. Hino, T. Kinoshita, and R. J. Bartlett, J. Chem. Phys. **121**, 1206 (2004).
[47] P. Dedíková, M. Pitoňák, P. Neogrády, I. Černušák, and M. Urban, J. Phys. Chem. A. **112**, 7115 (2008).
[48] M. Šulka, M. Pitoňák, P. Neogrády, and M. Urban, Int. J. Quantum Chem. **108**, 2159 (2008).
[49] A. Karton, P. R. Taylor, and J. M.L. Martin, J. Chem. Phys. **127**, 064104 (2007).
[50] For particular open-shell molecules the iterative triples contribution exceeds 0.7 mE_h if unrestricted HF orbitals are used, however, the contribution is smaller than this threshold with a restricted open-shell HF reference determinant even in these cases.
[51] MRCC, a string-based quantum chemical program suite written by M. Kállay. See also Ref. 17 as well as http://www.mrcc.hu/.
[52] CFOUR, a quantum chemical program package written by J. F. Stanton, J. Gauss, M. E. Harding, and P. G. Szalay with contributions from A. A. Auer, R. J. Bartlett, U. Benedikt, C. Berger, D. E. Bernholdt, Y. J. Bomble, O. Christiansen, M. Heckert, O. Heun, C. Huber, T.-C. Jagau, D. Jonsson, J. Jusélius, K. Klein, W. J. Lauderdale, D. A. Matthews, T. Metzroth, D. P. O'Neill, D. R. Price, E. Prochnow, K. Ruud, F. Schiffmann, S. Stopkowicz, J. Vázquez, F. Wang, and J. D. Watts and the integral packages MOLECULE (J. Almlöf and P. R. Taylor), PROPS (P. R. Taylor), ABACUS (T. Helgaker, H. J. Aa. Jensen, P. Jørgensen, and J. Olsen), and ECP routines by A. V. Mitin and C. van Wüllen. For the current version, see http://www.cfour.de.
[53] T. H. Dunning, Jr, J. Chem. Phys. **53**, 2829 (1970).
[54] M. E. Harding, J. Vázquez, J. Gauss, J. F. Stanton, and M. Kállay, "Towards highly accurate ab initio thermochemistry of larger systems: Benzene," J. Chem. Phys. (submitted).
[55] K. Kowalski and M. Valiev, J. Phys. Chem. A **110**, 13106 (2006).
[56] J. Csontos, Z. Rolik, S. Das, and M. Kállay, J. Phys. Chem. A **114**, 13093 (2010).
[57] B. Nagy, J. Csontos, M. Kállay, and G. Tasi, J. Phys. Chem. A **114**, 13213 (2010).
[58] P. Szakács, J. Csontos, S. Das and M. Kállay, "High-Accuracy Theoretical Thermochemistry of Atmospherically Important Nitrogen Oxide Derivatives," J. Phys. Chem. A (in press).
[59] M. Heckert, M. Kállay, and J. Gauss, Mol. Phys. **103**, 2109 (2005).
[60] S. A. Kucharski and R. J. Bartlett, J. Chem. Phys. **108**, 9221 (1998).

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